

IleoBond: A Mucoadhesive Therapeutic for Limited Small Bowel Crohn's Disease

Senior Design, Spring 2026

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Team Members



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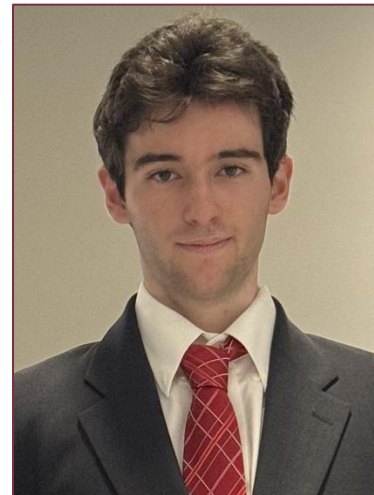
Alexander Moricette



Lazaro Castano



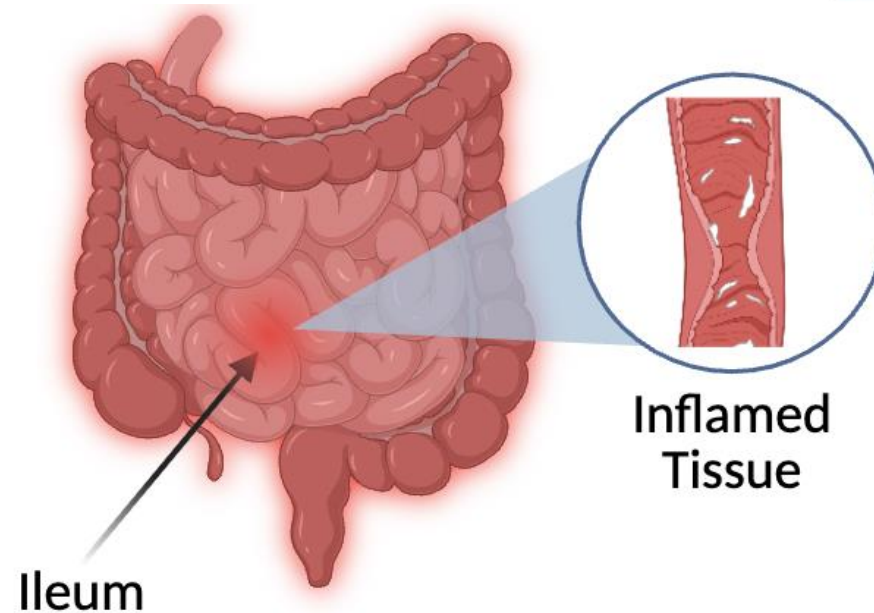
Eva Aguirre



Joshua Dindinger

Unmet Need in Crohn's Disease

- Poor drug localization in ileum
- Rapid transit limits exposure
- Systemic steroids lead to serious side effects



Needs Statement:

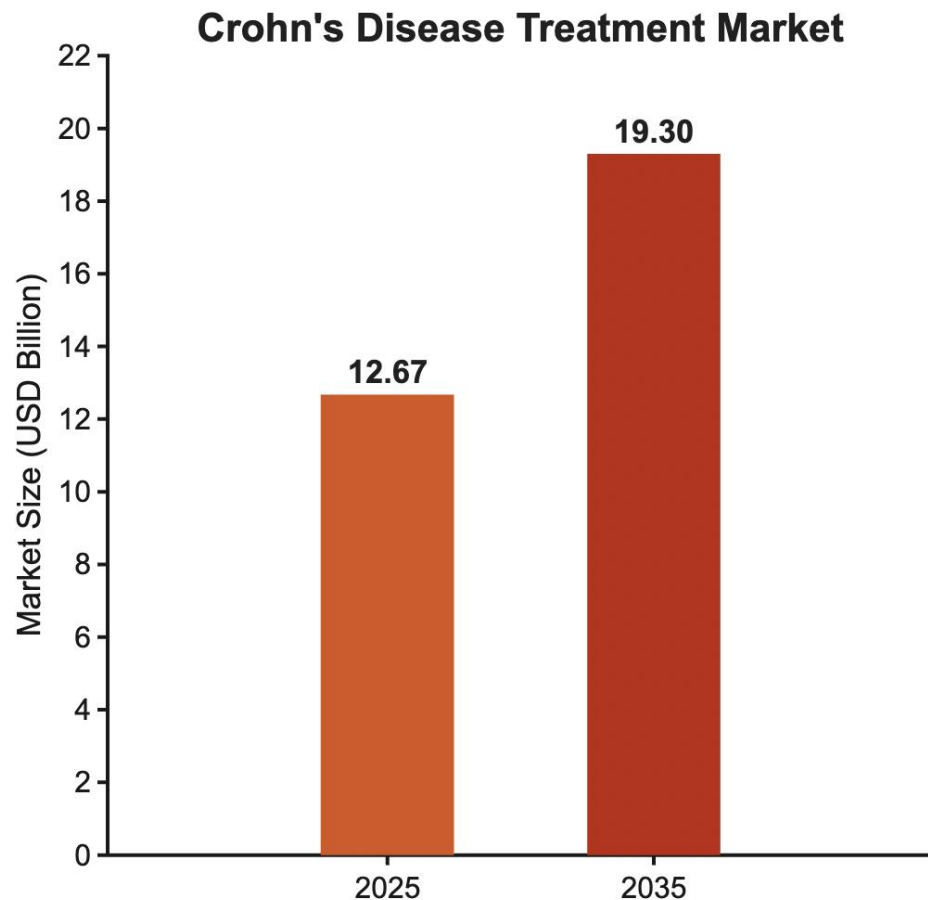
- A device that can anchor to ileal tissue and provide localized delivery of anti-inflammatory therapeutics while minimizing systemic exposure

Limitations of Current Treatments

Treatment	Side Effects
Oral Steroids	Non-specific delivery
Biologics	Costly, systemic risk
Enteric capsules	Inconsistent release
Surgery	Invasive, recurrence

Existing therapies treat the disease broadly, not precisely

Market Opportunity

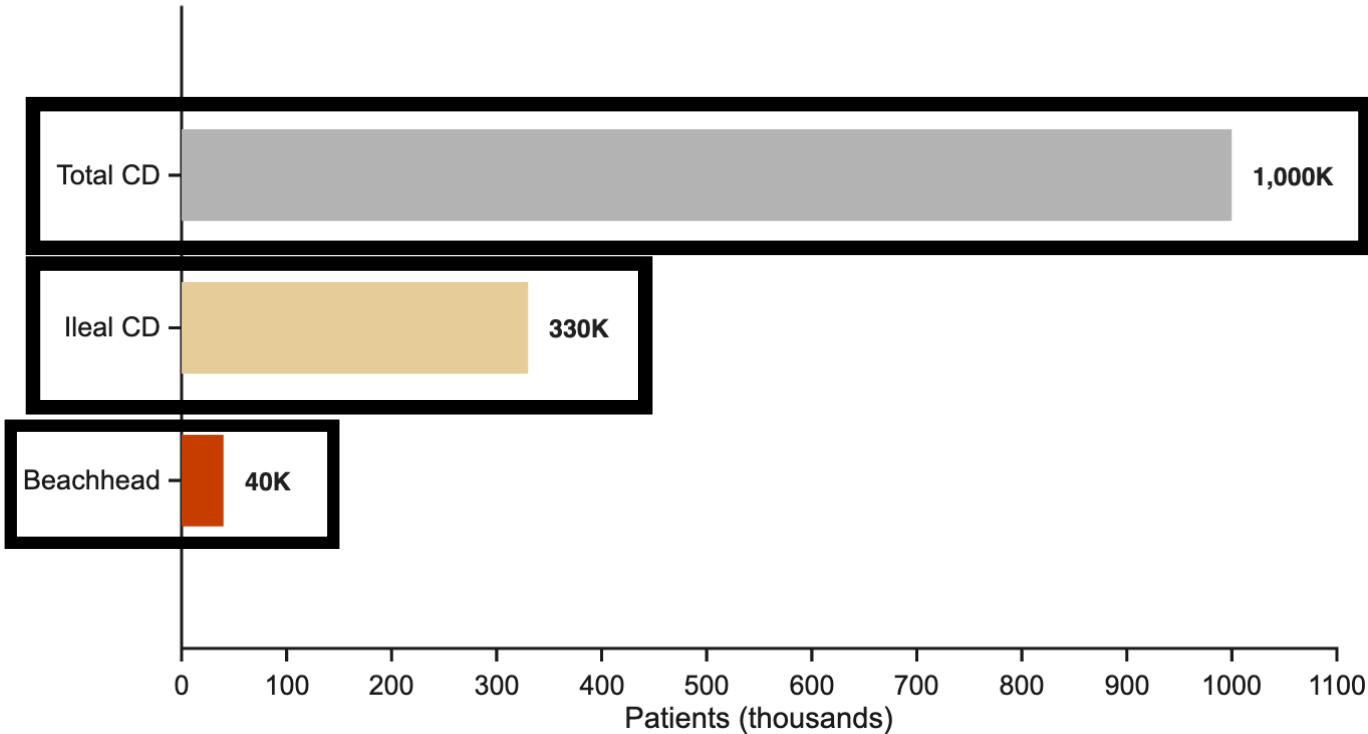


↑ 4.3%
Compound Annual
Growth Rate

- Multi billion-dollar treatment market with sustained growth
- Growth driven by biologics and targeted therapies
- Current therapies remain systemic, opportunity for localized delivery

Defined Entry Market for New Product

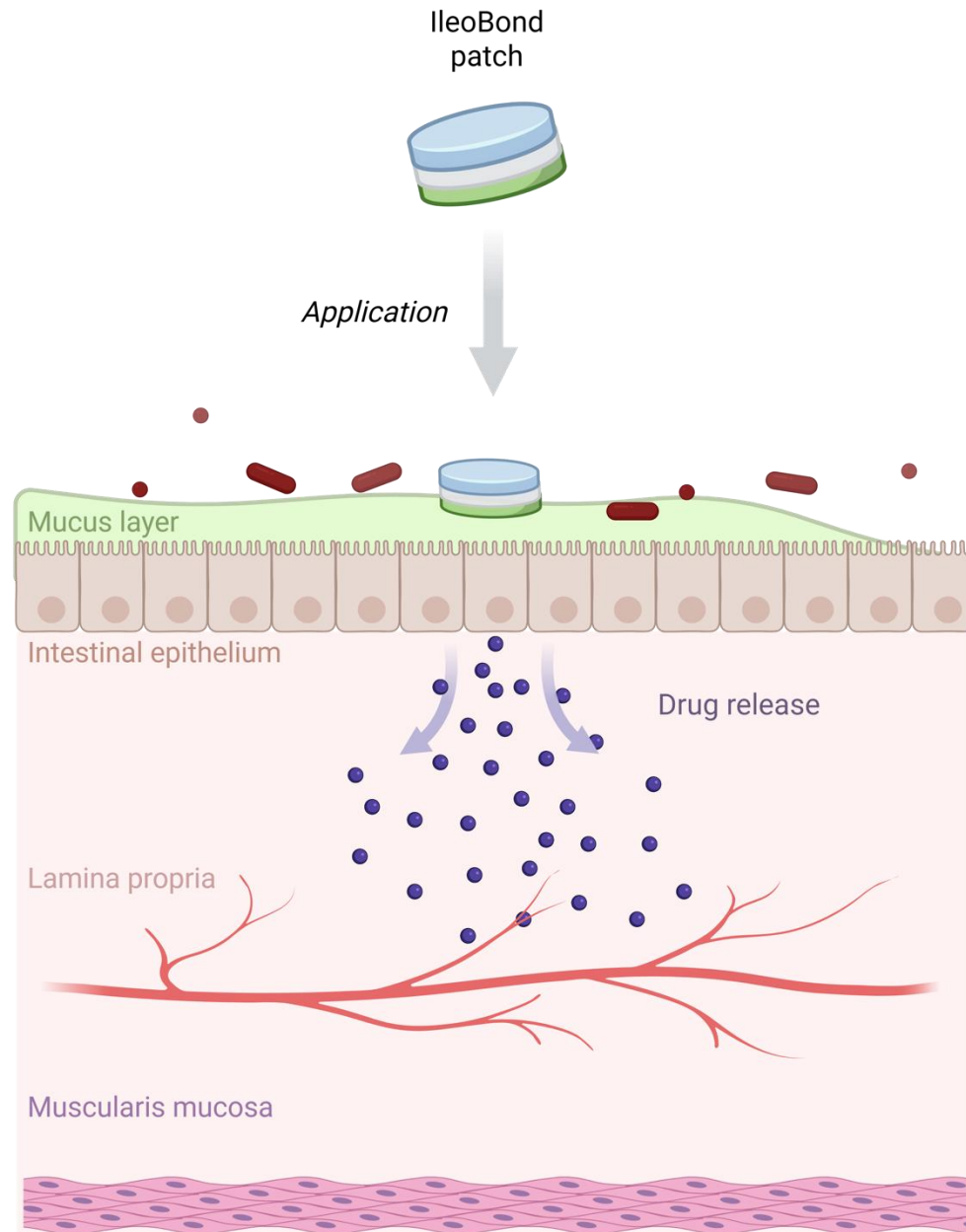
IleoBond Addressable Patient Funnel



- ~1M patients with Crohn's disease (CD) in the U.S.
- 30–40% present with ileal involvement

The initial addressable market focuses on moderate ileal CD patients with inadequate response to systemic therapies

IleoBond Patch Mode of Action



IleoBond patch layers



Backing

Drug

Adhesive

Device Design – Mucoadhesion

Why mucus?

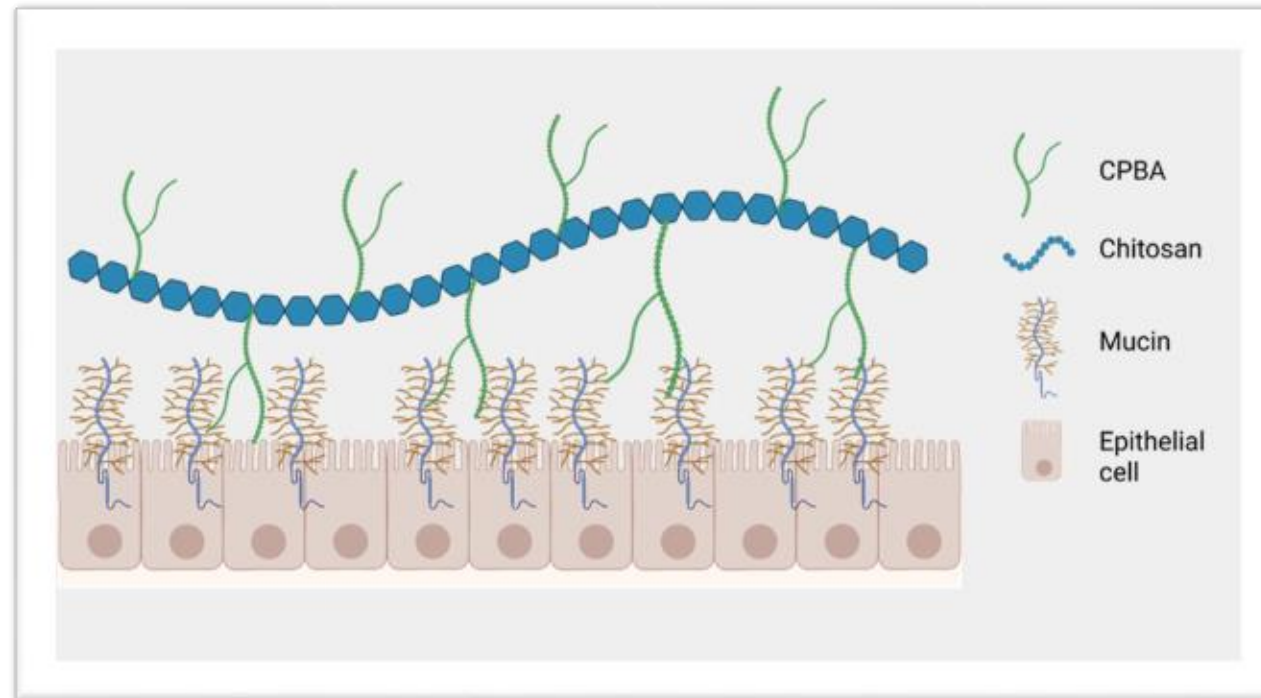
The intestine is lined by mucus polymers (~2000 ft² of surface area)

This layer governs the transport and retention of materials

Approach

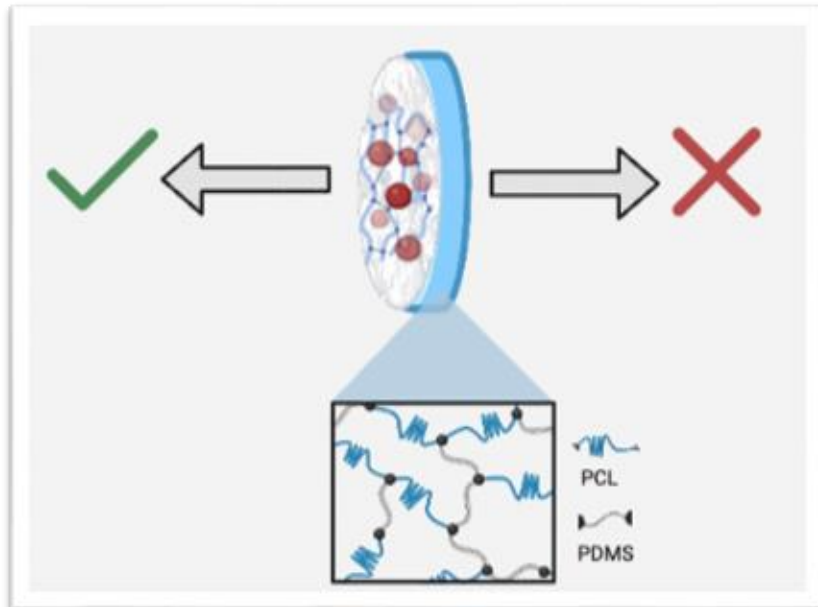
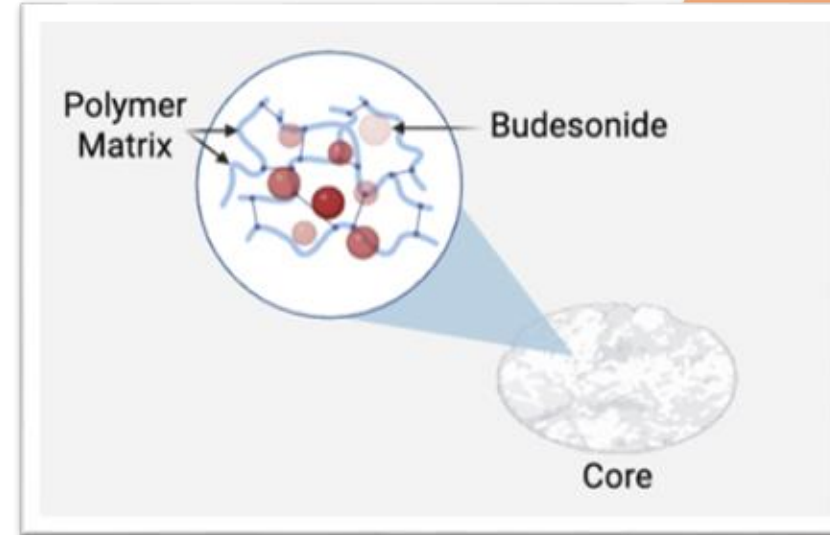
Adhesion layer composed of chitosan–CPBA polymer

A
mucoadhesive
intestinal
patch was
developed to
anchor to the
mucus layer



Device Design – Drug Reservoir & Backing

- Core composed of a polymeric matrix loaded with budesonide
 - Controlled drug release via swelling and diffusion



- Backing layer composed of hydrophobic PCL/PDMS composite
 - Acts as a barrier to prevent drug loss into the lumen

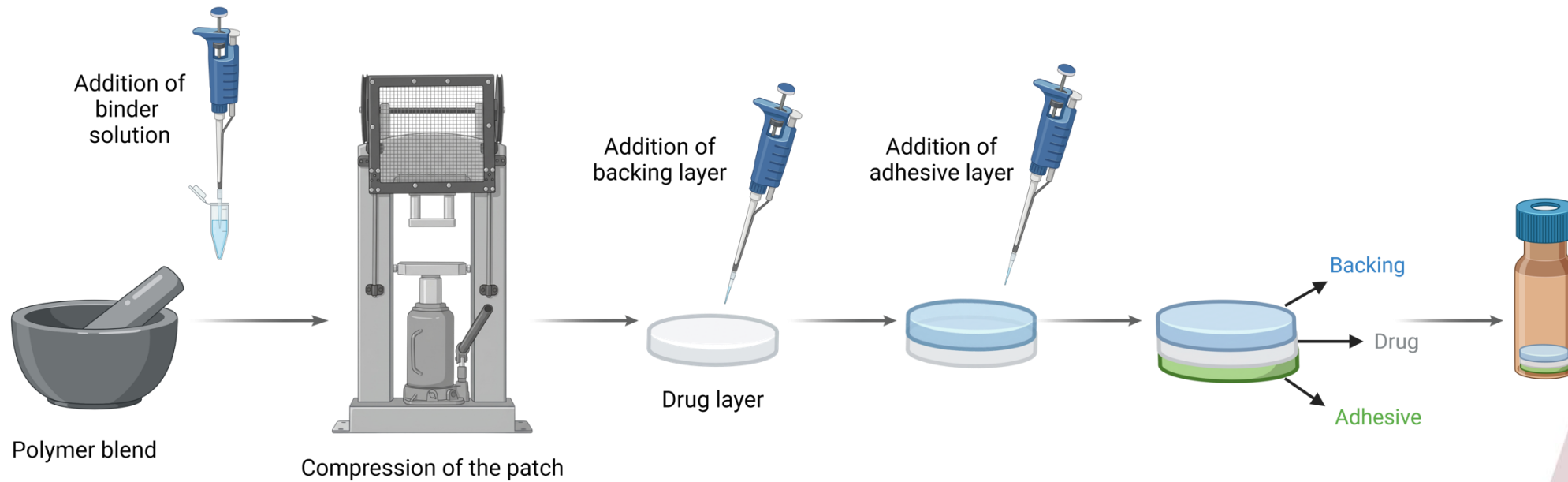


Clinical Need into Design Inputs

Parameter	Target
Geometry	8 mm × 8 mm × 0.8 mm
Drug Loading	6 wt.% budesonide
Release Rate	3 mg/day
Retention Time	8–12 weeks
Target Location	Terminal ileum (± 12 cm from ileocecal valve)
Degradation	Biodegradable polymers

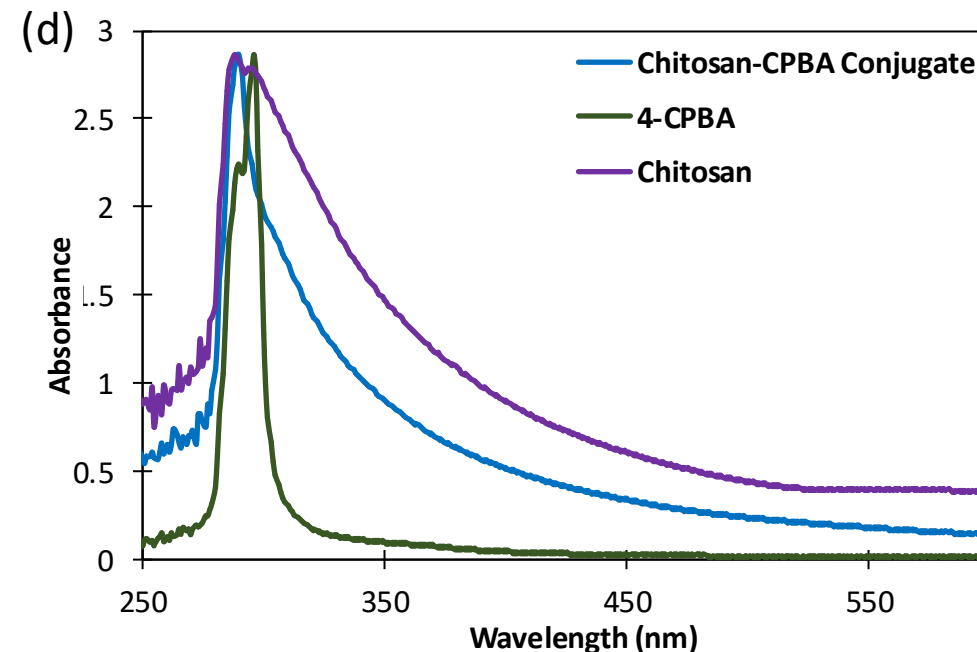
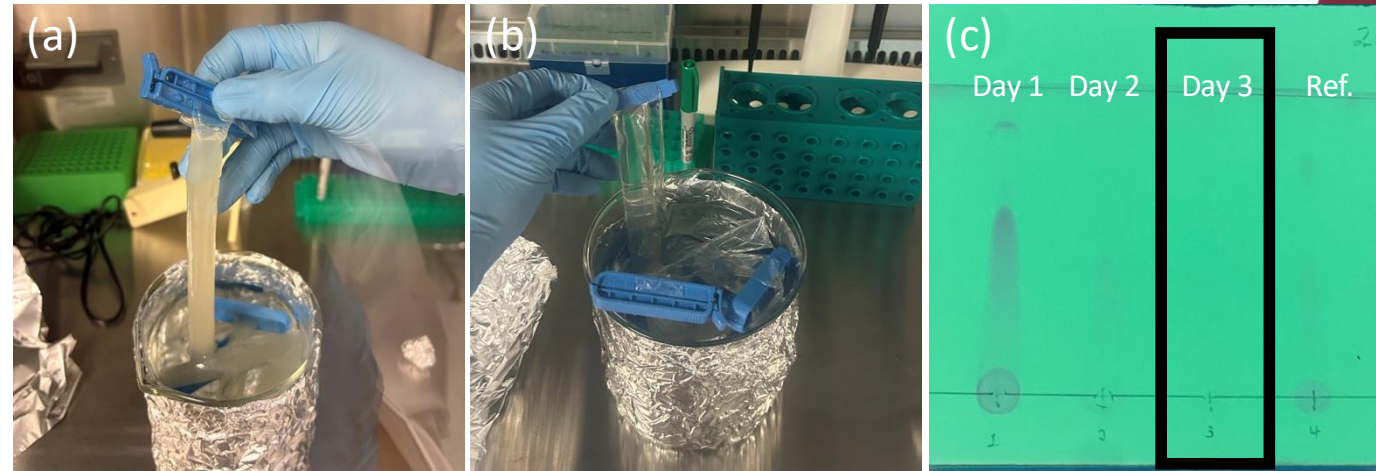
- Enables localized drug delivery to the ileocecal region, maintaining retention under peristalsis while functioning effectively within mucus-rich GI environment

Fabrication Workflow



Mucoadhesive Polymer

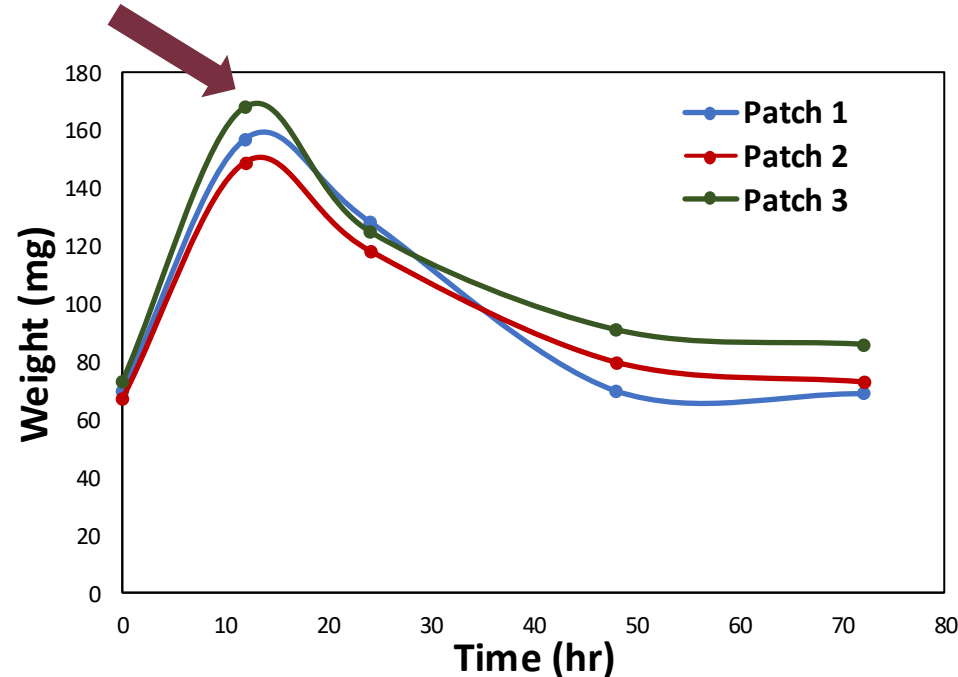
- CBPA-chitosan conjugate synthesized
- Unreacted CPBA separated by dialysis and confirmed by thin-layer chromatography
- UV-Vis spectroscopy shows peak positions consistent with incorporation of the CPBA into the chitosan backbone



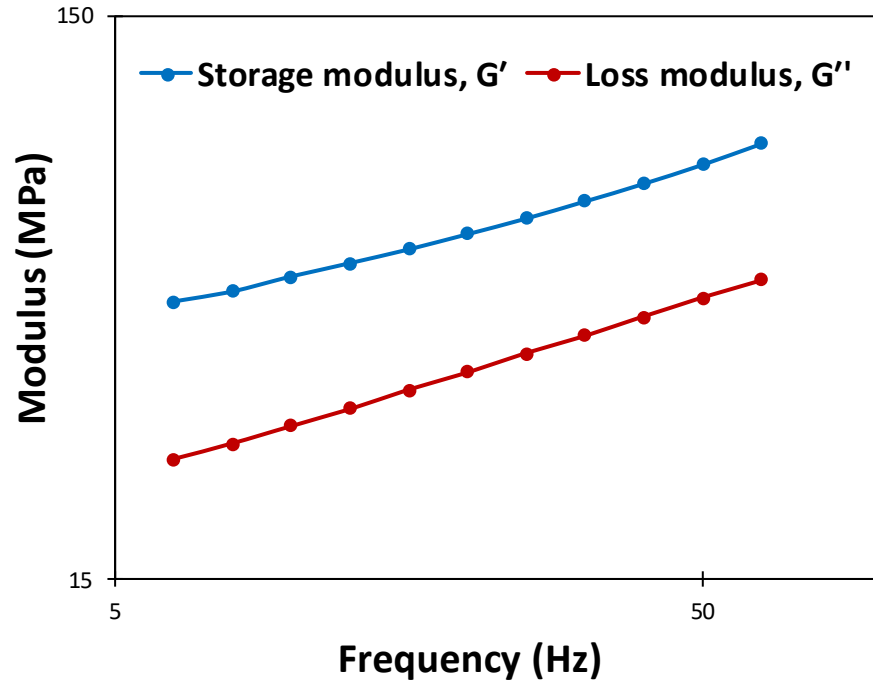
Swelling Behavior

- Swelling facilitates diffusion-driven release of budesonide from the polymer matrix
- Hydration with 120-130% mass increase within ~12 hours
- Gradual deswelling indicates controlled fluid release and matrix relaxation over time

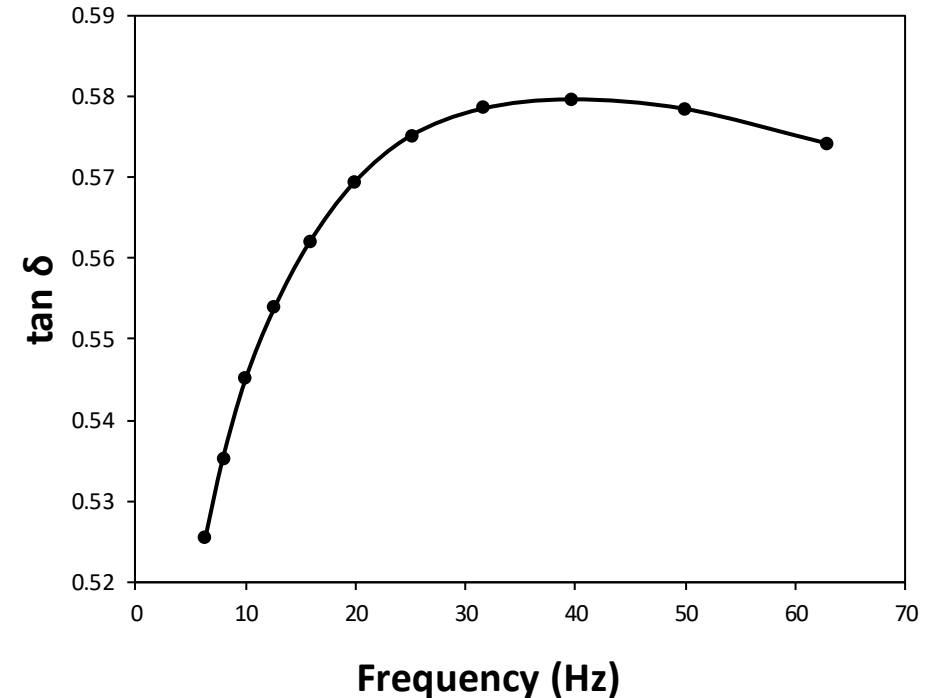
Patch	Peak (mg)	% Swelling	AUC (mg·h)	Swelling Rate (mg/h)	Deswelling Rate (mg/h)
1	157	124%	7116	7.25	-1.47
2	149	122%	7109	6.83	-1.27
3	168	130%	7920	7.92	-1.37



Mechanical Characterization



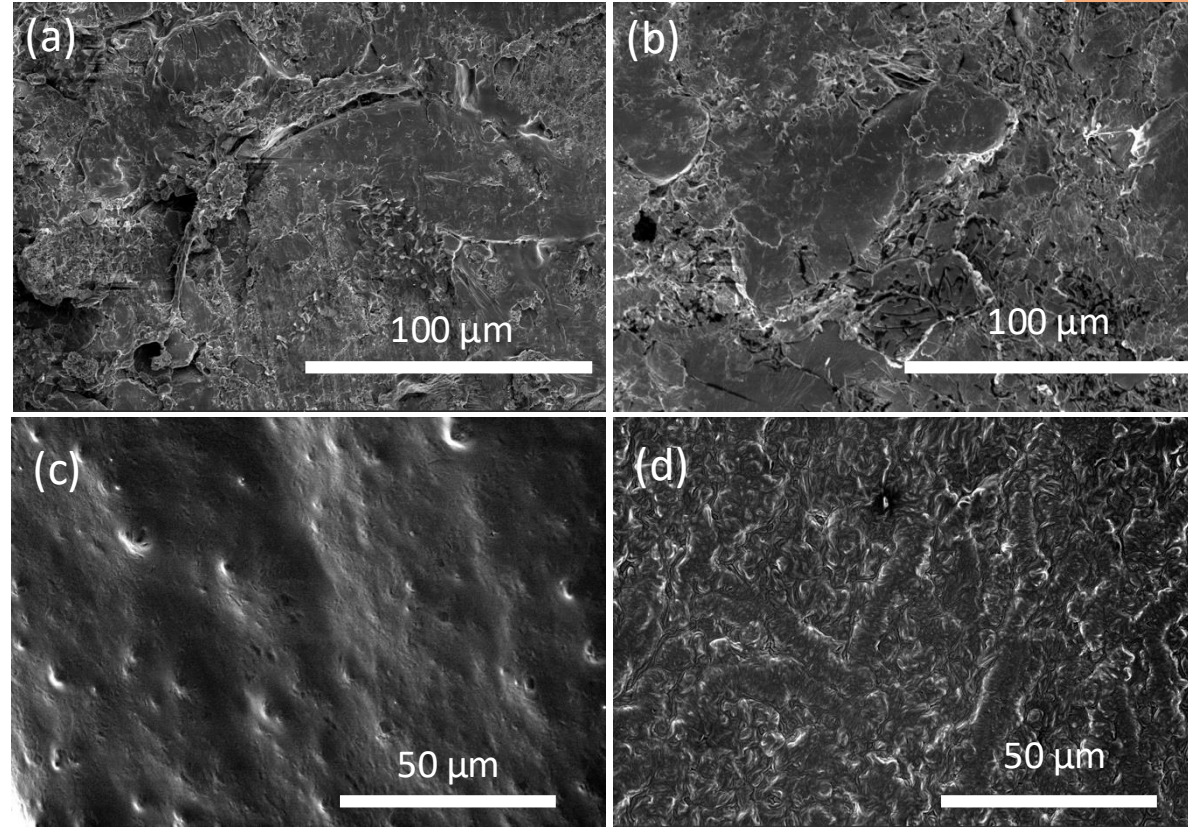
- Storage and loss moduli follow a power law scaling,
 - $G'(f) = 27.1 f^{0.28}$ MPa
 - $G''(f) = 13.5 f^{0.32}$ MPa
- $G' > G''$ across all frequencies
 - Predominantly *elastic behavior*



- $\tan \delta < 1$ (0.525–0.580); mean 0.562 ± 0.018
 - Confirms elastic-dominated viscoelastic behavior
- Core structure supports mechanical stability under peristaltic conditions

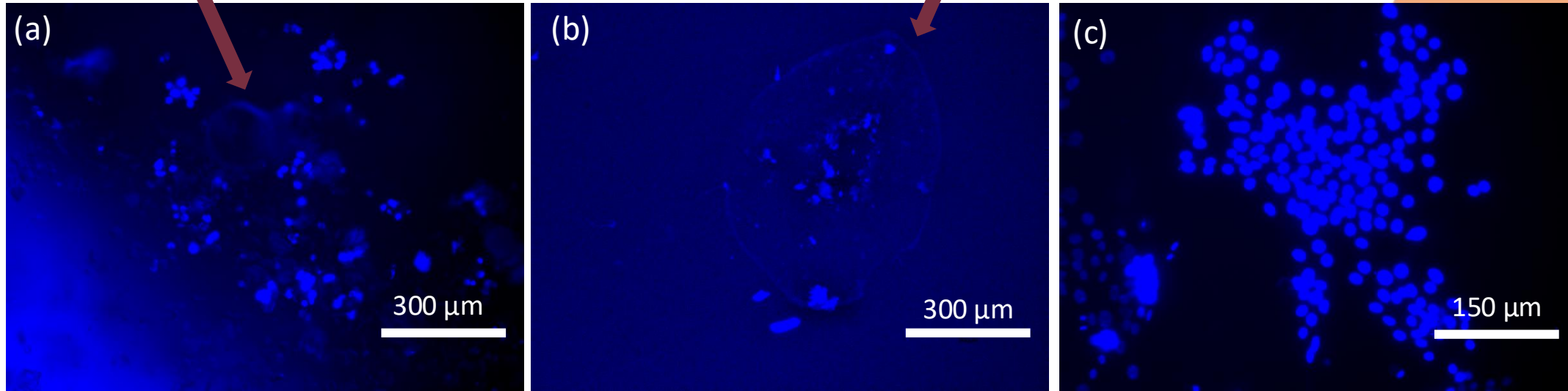
Microstructural Features

- Scanning electron microscopy (SEM) revealed layer-specific morphologies
- Core: heterogeneous matrix
- Backing: continuous hydrophobic surface
- Adhesive: textured interface
- Structure aligns with controlled release, barrier function, and mucoadhesion



SEM micrographs of (a) loaded drug reservoir core; (b) unloaded core; (c) backing layer surface; (d) mucoadhesive layer.

In Vitro Compatibility

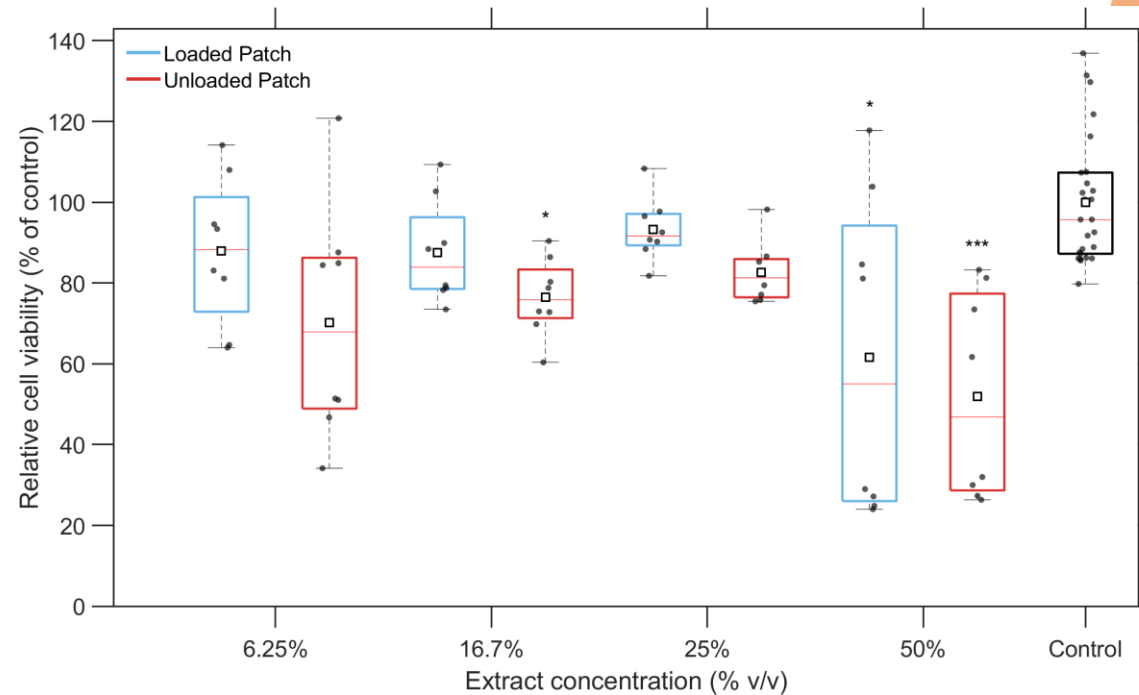


(a) Culture exposed to unloaded patch. (b) Culture exposed to drug-loaded patch. (c) Control culture.

- NIH3T3 fibroblasts used to assess material biocompatibility following 48 h exposure
- Both unloaded and drug-loaded patches support cell attachment and normal nuclear morphology
- These results suggest baseline cytocompatibility

Metabolic Activity Assessment

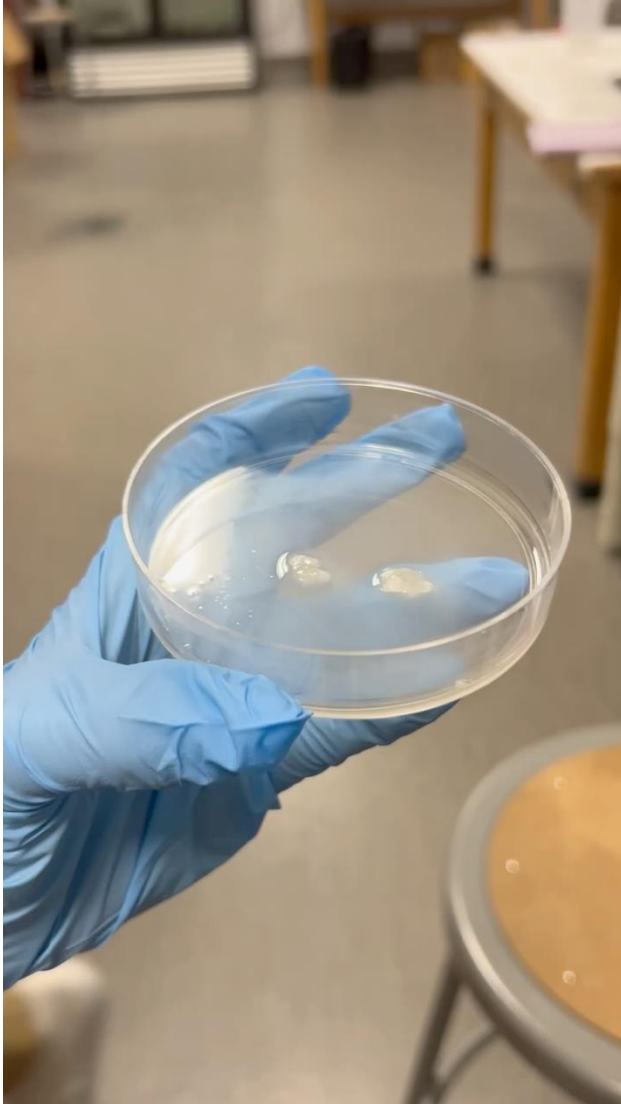
- Patch extracts influence metabolic activity (Kruskal–Wallis $p = 1.83 \times 10^{-5}$)
- Drug loaded patches did not exhibit greater cytotoxicity than unloaded patches at any concentration



(* $p < 0.05$, *** $p < 0.001$)

- At moderate extract concentrations, loaded patches were not significantly different from control ($p = 0.995, 0.942$)
- Drug loaded system does not introduce overt *in vitro* cytotoxicity

Attachment to Simulated Mucosal Surface

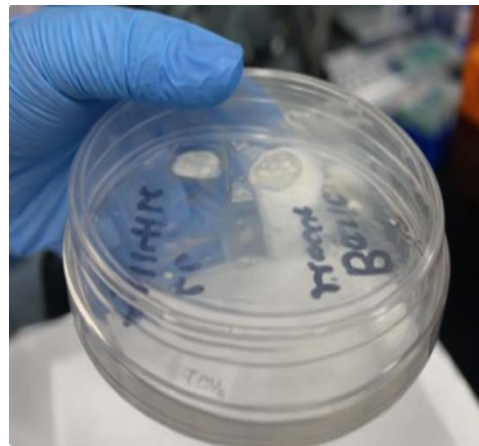


Test Conditions

- Synthetic mucus composed of porcine mucin and simulated intestinal fluid (pH = 7.4)
- Patch placed on mucus surface and gently agitated
- Adhesion was steady over 2 days

Observation

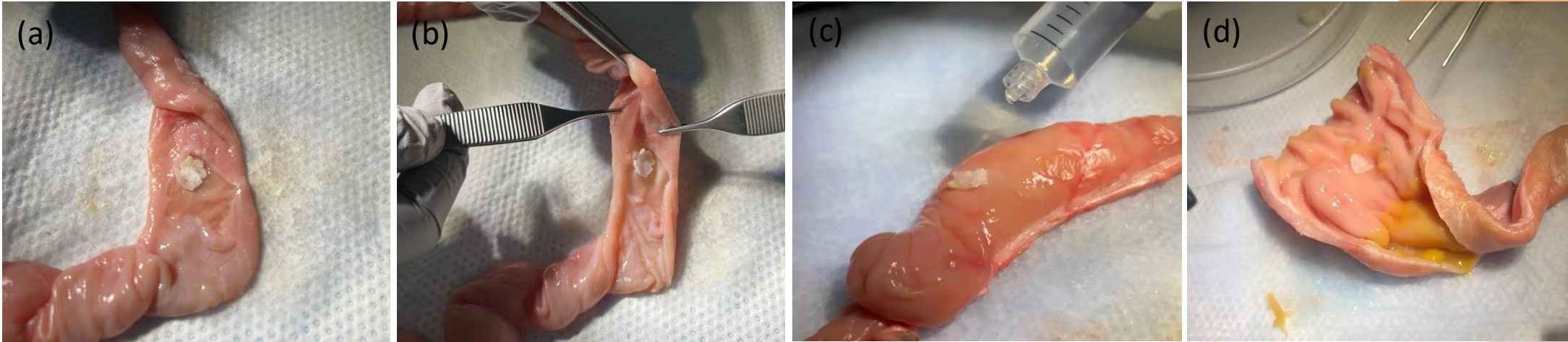
- Patch remained adhered without displacement
- No visible detachment or sliding



Patches demonstrated cohesive retention to the petri dish surface upon inversion



Porcine Intestine Adhesion

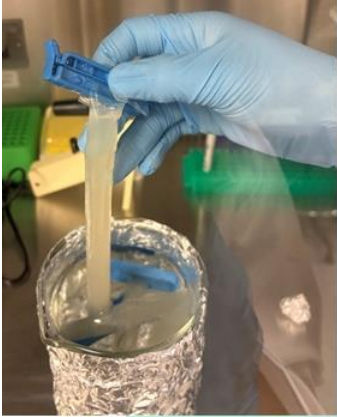


(a) Patch placed on mucosa to assess initial attachment; (b) Mechanical disturbance applied to evaluate retention; (c) Fluid introduced to simulate peristaltic flow; (d) Patch tested on folded intestinal tissue.

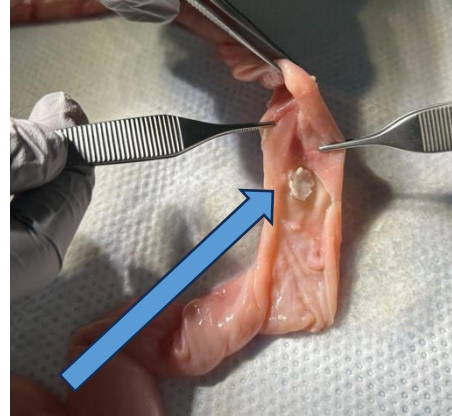
- Attachment was influenced by application sequence, tissue hydration, and surface contact
- Mucoadhesive performance decreased when adhesive layer was dried prior to application
- Observations prompted refinement of patch design and application approach

Current Design and Change History

*Development of
Mucoadhesive
(CBPA-Chitosan
conjugate)*



December '25-
January '26



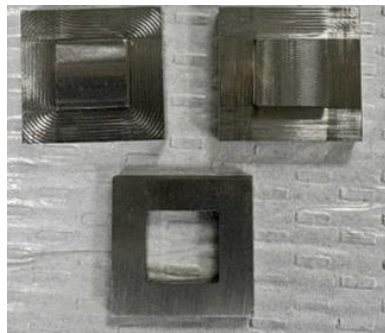
February-
March 26'
*Modification into a circle,
optimizing manufacturing +
storage conditions*

November '25

*Assembly of first
patch (square)*



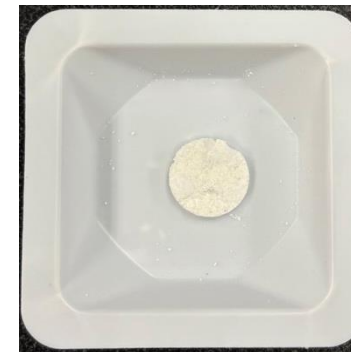
Patch core



Hydraulic press mold

January 16th,
2026

*Visit to Mayo
Clinic, adhesive
observations*



Circular core



Cylinder
'punchers'

Manufacturing Considerations

- Statistical tests run on patches from three different batches of cores

Unloaded:

Batch 1 & 2

Loaded:

Batch 3

- Compared batch to batch variability in terms of diameter, thickness, and weight

Mann-Whitney U Test

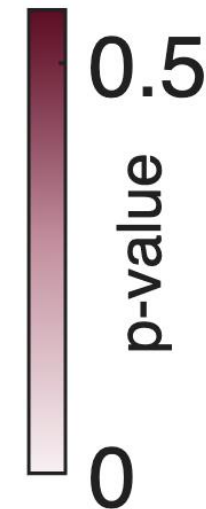
1 vs 3	0.018	0.141	0.0017
2 vs 3	0.179	0.277	0.565
1 vs 2	0.277	0.084	0.0017

Weight
Thickness
Diameter

Unequal Variance t-Test

1 vs 3	0.011	0.168	0.0038
2 vs 3	0.403	0.168	0.447
1 vs 2	0.212	0.047	0.0068

Weight
Thickness
Diameter



Budesonide loading did not significantly affect core geometry

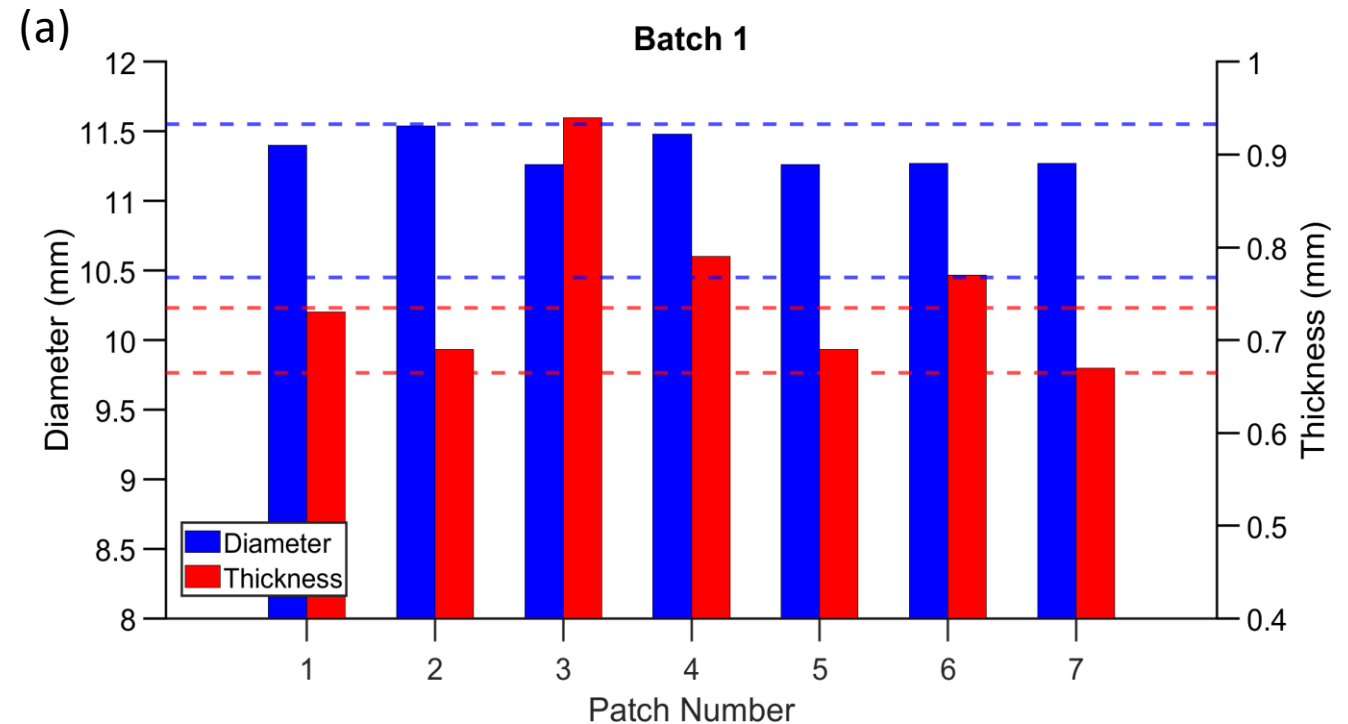
Improving Manufacturing Workflow

- Possible sources of variability:

1. Inconsistent mold filing
2. Variability in press duration and force applied, removal of patch

- Proposed improvements:

1. Consistent patch removal
2. Prevent sticking of patch to mold
 - a. Increase magnesium stearate
 - b. Apply non-stick coating to prevent adhesion to mold
3. Establish batch tolerances for consistency
 - a. Figure (a) depicts example batch 1

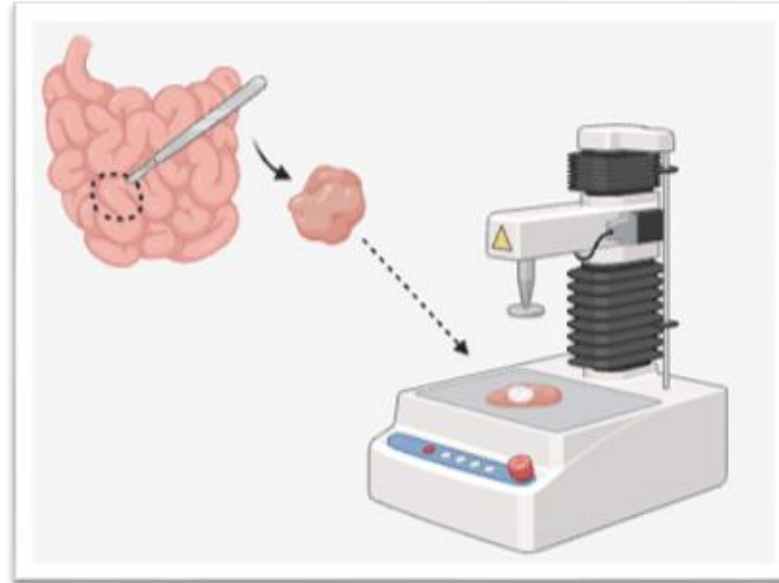


Dashed lines indicating $\pm 5\%$ (blue) and $\pm 5\%$ (red) tolerance ranges used to assess dimensional consistency

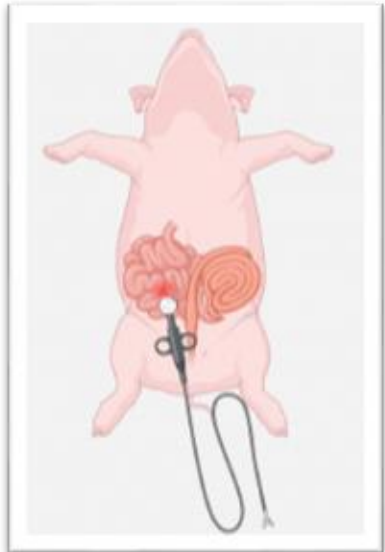
Future Testing

Adhesion Validation

Repeat of *ex vivo* testing on fresh intestinal tissue & force-based detachment analysis using specialized adhesion testing instruments.



Large in vivo model studies

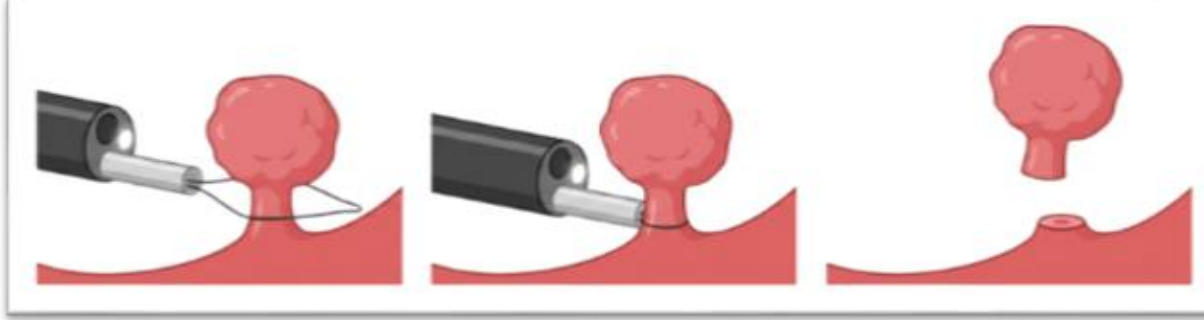


Long-term
cytotoxicity
profiling to
assess
systemic safety

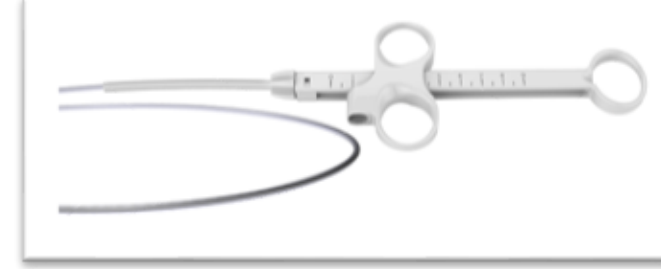
Tissue
permeation
studies to
quantify drug
uptake

Retention time
quantification
within a large
mammal GI
tract model

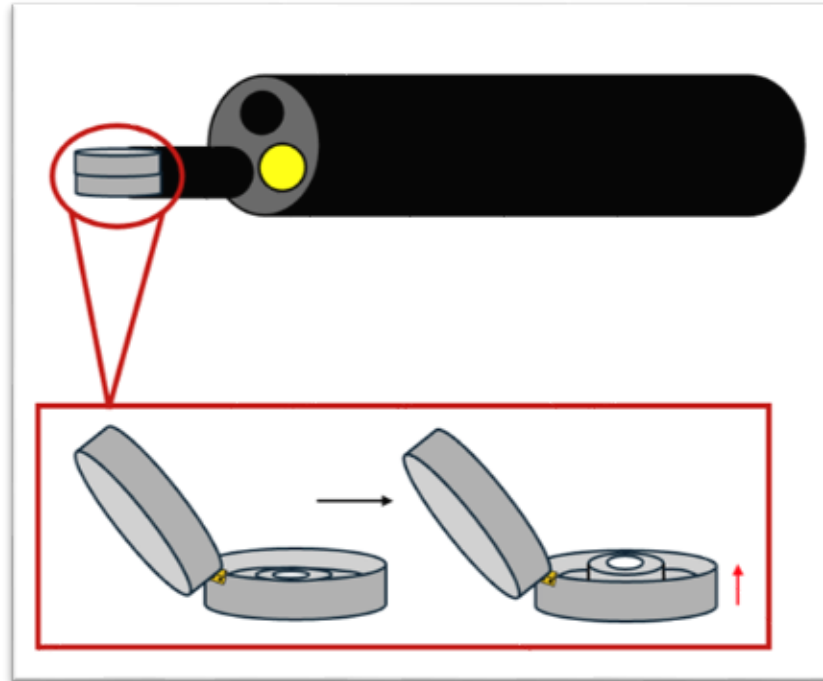
IleoBond Applicator



Polypectomy snare mechanism of action



Scissor handle used to actuate snare opening and closure



IleoBond Applicator:

- Fits standard colonoscope instrument channel
- Scissor actuation opens capsule and advances patch loaded stage
- Patch pre-loaded mucoadhesive side up for direct mucosal contact
- Sustained pressure ensures conformal adhesion over target ulceration

Intellectual Property

Novel Features

- Mucoadhesive intestinal patch forming a retained ileal drug depot
- Local budesonide delivery independent of GI transit
- Polymer system tuned for adhesion, swelling, and controlled release

Potential Patent Categories

- Drug delivery mechanism (multilayer reservoir & unidirectional backing)
- Controlled release formulation
- Method of treatment for limited small-bowel Crohn's disease

IP Strategy

- Provisional covering composition, structure, anchoring, and use
- Trade secrets for polymer ratios, fabrication parameters, and release tuning

In Conclusion...



Addresses a clinical gap with localized, transit-independent LSBCD delivery

Demonstrates structural integrity, baseline cytocompatibility, and preliminary adhesion

However:

All experimental findings remain preclinical with no *in vivo* validation

Prioritize adhesion and *in vivo* studies to establish safety and efficacy

Standardize manufacturing and define a clear regulatory path

Acknowledgements



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Dr. Stephen Arce
Dr. Emily Hubicki



IleoBond FAQs

- *What is the drug loading in IleoBond?*

The reservoir contains ~6 wt.% budesonide within a multicomponent polymer matrix.

- *What specific materials make up the patch?*

Core: Eudragit RS100, HPMC K100M, Carbopol 934P, PCL, PLA, PEG-400;
Adhesive: chitosan–CPBA; Backing: PCL/PDMS.

- *How is the patch eventually cleared?*

The system is fully biodegradable, with degradation over ~8–12 weeks in the intestinal environment.

- *How does inflammation affect performance?*

Inflammation alters pH, permeability, and transport, which could impact adhesion and diffusion. This remains to be validated experimentally.

Comparison of Group Differences: U & T Statistics

Comparison	Weight (W)		Thickness (T)		Diameter (D)	
	U	t	U	t	U	t
1 vs. 3	6	3.123	13	1.499	0	4.239
2 vs. 3	14	0.881	16	1.499	20	-0.791
1 vs. 2	16	1.323	11	-2.341	0	3.787

Core: Polymer Ratios

Wet Mass (mg): 75 mg

Core Polymers	Percentages (%)	Amount of Polymer (mg)
Budesonide	6	4.5
Carbopol 934P	12	9
PVP K90	12	9
Lactose Monohydrate	20	15
PEG 400	12	9
Magnesium Stearate	1	.75
HPMC K100M	16	12
PCL	5	3.75
Eudragit RS100	5	3.75
PLA	6	4.5

MTT Assay Principle

- Trypan blue staining is simple but **insensitive/low throughput method** of evaluating cell membrane integrity (cell proliferation or death).
- Assessment of radioactive substance uptake (e.g. tritium-labeled thymidine) is **time consuming and requires handling of radioactive substances**.
- MTT (3-(4,5-dimethylthiazol-2-yl)-2, 5-diphenyltetrazolium bromide) is **reduced to purple colored formazan** in the living cells.
- The absorbance of this colored solution can be **quantified by measuring at a certain wavelength** (550-600 nm) by spectrophotometer.
- This reaction only takes place **only when mitochondrial reductase enzymes are active**, which is **directly related to the number of viable cells**.
- Often employed as a **cytotoxicity assay on dose-dependent manner**.

